

AMATH 353 HW 7

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Question 1: An Easier Shock Solution

One setting where the method of characteristics fails is when the characteristics intersect and the solution becomes infinitely steep. This leads to the formation of *shock wave*, and we saw that the shock moves along a characteristic $(x_s(t), t)$ which can be determined from Rankine-Hugoniot condition

$$\frac{dx_s(t)}{dt} = \frac{\phi(u^+) - \phi(u^-)}{u^+ - u^-}.$$

Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + uu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 2, & x < -1, \\ 1 - x, & |x| \leq 1. \\ 0, & x > 1. \end{cases}$$
